Souhrid Mukherjee, PhD

Computational Biochemist and Data Scientist

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RESEARCH SUMMARY

Computational biologist with 13 years of research experience, 10 of them computational, working on the molecular mechanisms of rare and complex disease and on the design and safety assessment of the drugs that treat them. First author on three of 17 peer-reviewed publications, spanning human genetics, molecular simulation, cheminformatics, multi-omics and tissue engineering. Most of this work has been collaborative, usually as the computational scientist in a team of experimentalists and clinicians.

APPOINTMENTS

Basic Life Sciences Research Scientist

Stanford Cardiovascular Institute, Palo Alto, CA | May 2025 to Present

Full-time staff scientist appointment, not a postdoctoral position. Visiting Scientist, 2022 to 2024.

- Build machine learning models of drug response and toxicity that combine whole-genome sequencing, transcriptomics and functional phenotyping of human iPSC-derived cardiomyocytes.
- Built the end-to-end genome analysis pipeline for the institute iPSC biobank: sample identity, sequencing quality, genetic ancestry, and pathogenic variant annotation.
- Developing a personalized cardiotoxicity atlas that predicts drug-induced cardiotoxicity from interactions between drugs and patient genetic variants.
- Ran the molecular dynamics for the lab cardiac fibrosis drug studies, including validation that saracatinib binds Src kinase more strongly and more stably than four other candidates over 100 ns.
- Mentor junior researchers and translate *in vitro* cellular phenotypes into clinical endpoints.

Senior Scientist, AI and Computational Biology

Greenstone Biosciences, Palo Alto, CA | July 2022 to Present

Full-time July 2022 to May 2025, as Scientist then Senior Scientist. Part-time employee in the same Senior Scientist role since May 2025.

- Co-developed ADMET-AI, an open platform predicting 41 ADMET properties fast enough to triage large chemical libraries, with the Zou group at Stanford.
- Led conceptualization and deployment of ADMET-AI DICT for drug-induced cardiotoxicity. Trained on 555 FDA DICTRank drugs, it reached ROC AUC 0.72 against 0.67 for SwissADME, and SHAP analysis identified CYP2D6 substrate status and Nrf2 antioxidant response as the strongest drivers.
- Designed a high-throughput computational discovery platform spanning screening, optimization and molecular dynamics of protein and ligand binding.
- Combine generative design using SyntheMol over Enamine REAL building blocks with QSAR and ADMET scoring in an active learning loop, optimizing potency against BBB permeability and hERG liability.
- Built a target identification platform over public and proprietary multi-omics data, and a focused knowledge graph of known and predicted drug and disease relationships.
- Delivered a 2,156-compound ADMET report for a combinatorial analogue series across four modification domains.
- Led a team on a contract research project for a non-profit patient foundation, modeling Alexander disease in patient-derived iPSC lines, and performed the differentiation, culture and imaging of neural progenitors and astrocytes.
- Partner daily with medicinal chemists and external CRO partners to move ranked compounds into synthesis.

Graduate Researcher, Computational Biology

Vanderbilt University, Nashville, TN | August 2016 to June 2022

Co-mentored by Tony Capra, PhD and Jens Meiler, PhD, Department of Biological Sciences.

- Developed DiGePred, a random forest predictor of digenic disease gene pairs, with the Undiagnosed Diseases Network. Applied to more than 100 undiagnosed patients and several rare disease cohorts, and it contributed a candidate gene pair for Mayer-Rokitansky-Küster-Hauser syndrome.
- Led an investigation across clinicians, biochemists and electrophysiologists into two neighboring *de novo KCNC2* variants causing developmental epileptic encephalopathy. Molecular dynamics and free-energy analysis predicted V471L as gain of function and V469L as loss of function; biochemistry and electrophysiology confirmed both.

Junior Research Fellow

Institute for Stem Cell Biology and Regenerative Medicine (inStem) and NCBS, Bengaluru, India | 2014 to 2016

Muscle protein biochemistry laboratory of John Mercer, PhD and Jim Spudich, PhD.

- Purified individual sarcomeric proteins, reconstituted cardiac thin filaments *in vitro*, and benchmarked functional assays on tropomyosin and troponin T mutations, predicting which protein domains bias toward hypertrophic or dilated cardiomyopathy.

Earlier research training

Research Intern, Tissue Engineering Group, Indian Institute of Technology Delhi, 2014

Research Intern, Stem Cell Facility, All India Institute of Medical Sciences, New Delhi, 2013

Summer Research Intern, Bose Institute, Kolkata, 2012. First training in molecular dynamics simulation.

EDUCATION

Doctor of Philosophy, Biological Sciences (Computational Biology)

Vanderbilt University, Nashville, TN | 08/2016 to 06/2022

Master of Science, Biochemistry

University of Calcutta, Kolkata, India | 07/2011 to 06/2013

Bachelor of Science, Biochemistry
Presidency College, Kolkata, India | 07/2008 to 06/2011

PUBLICATIONS

17 peer-reviewed publications, three as first author. The full record is on Google Scholar and ORCID.

First author

- **Mukherjee S**, **Swanson K**, **Walther P**, **Shivnaraine RV**, **Leitz J**, **Pang PD**, **Zou J**, **Wu JC**. *ADMET-AI enables interpretable predictions of drug-induced cardiotoxicity.* Circulation. 2025;151(3):285-287. PMC11996029.
- **Mukherjee S**, **Cassini TA**, **Hu N**, **Yang T**, **Li B**, **Shen W**, **Moth CW**, **Rinker DC**, **Sheehan JH**, **Cogan JD**, **Newman JH**, **Hamid R**, **Macdonald RL**, **Roden DM**, **Meiler J**, **Kuenze G**, **Phillips JA**, **Capra JA**. *Personalized structural biology reveals the molecular mechanisms underlying heterogeneous epileptic phenotypes caused by de novo KCNC2 variants.* HGG Advances. 2022;3(4):100131. PMC9399384.
- **Mukherjee S**, **Cogan JD**, **Newman JH**, **Phillips JA 3rd**, **Hamid R**, **Meiler J**, **Capra JA**. *Identifying digenic disease genes via machine learning in the Undiagnosed Diseases Network.* American Journal of Human Genetics. 2021;108(10):1946-1963. PMC8546038.

Contributing author

- **Cho S**, **Rhee S**, **Madl CM**, **Caudal A**, **Thomas D**, **Kim H**, **Kojic A**, **Shin HS**, **Mahajan A**, **Jahng JW**, **Wang X**, **Thai PN**, **Paik DT**, **Wang M**, **Mullen M**, **Baker NM**, **Leitz J**, **Mukherjee S**, **Winn VD**, **Woo YJ**, **Blau HM**, **Wu JC**. *Selective inhibition of stromal mechanosensing suppresses cardiac fibrosis.* Nature. 2025;642(8068):766-775. PMC12176515.
- **Zhang H**, **Thai PN**, **Shivnaraine RV**, **Ren L**, **Wu X**, **Siepe DH**, **Liu Y**, **Tu C**, **Shin HS**, **Caudal A**, **Mukherjee S**, **Leitz J**, **Wen WTL**, **Liu W**, **Zhu W**, **Chiamvimonvat N**, **Wu JC**. *Multiscale drug screening for cardiac fibrosis identifies MD2 as a therapeutic target.* Cell. 2024;187(25):7143-7163.e22. PMC11645214.
- **Zhang H**, **Ren L**, **Huang R**, **Tan R**, **Solow-Cordero DE**, **Mukherjee S**, **Wu JC**. *New approach methodologies: a drug screening platform to identify antifibrotic compounds.* Nature Protocols. 2026.
- **Swanson K**, **Walther P**, **Leitz J**, **Mukherjee S**, **Wu JC**, **Shivnaraine RV**, **Zou J**. *ADMET-AI: a machine learning ADMET platform for evaluation of large-scale chemical libraries.* Bioinformatics. 2024;40(7). PMC11226862.
- **Patra D**, **Sayed IM**, **Mukherjee S**, **Piplani H**, **Habtezion A**, **Rosen MJ**, **Wu JC**. *Integrating human intestinal organoids into the FDA New Approach Methodologies for drug discovery.* Advanced Science. 2026;13(20):e22276. PMC13067788.
- **Zushin PH**, **Mukherjee S**, **Wu JC**. *FDA Modernization Act 2.0: transitioning beyond animal models with human cells, organoids, and AI/ML-based approaches.* Journal of Clinical Investigation. 2023;133(21). PMC10617761.
- **Guo H**, **Yu X**, **Liu Y**, **Paik DT**, **Justesen JM**, **Chandy M**, **Jahng JWS**, **Zhang T**, **Wu W**, **Rwere F**, **Zhao SR**, **Pokhrel S**, **Shivnaraine RV**, **Mukherjee S**, **Simon DJ**, **Manhas A**, **Zhang A**, **Chen CH**, **Rivas MA**, **Gross ER**, **Mochly-Rosen D**, **Wu JC**. *SGLT2 inhibitor ameliorates endothelial dysfunction associated with the common ALDH2 alcohol flushing variant.* Science Translational Medicine. 2023;15(680):eabp9952. PMC10297796.
- **Mikhael S**, **Dugar S**, **Morton M**, **Chorich LP**, **Tam KB**, **Lossie AC**, **Kim HG**, **Knight J**, **Taylor HS**, **Mukherjee S**, **Capra JA**, **Phillips JA 3rd**, **Friez M**, **Layman LC**. *Genetics of agenesis and hypoplasia of the uterus and vagina: narrowing down the number of candidate genes for Mayer-Rokitansky-Küster-Hauser syndrome.* Human Genetics. 2021;140(4):667-680. PMC9211441.
- **Gangadharan B**, **Sunitha MS**, **Mukherjee S**, **Chowdhury RR**, **Haque F**, **Sekar N**, **Sowdhamini R**, **Spudich JA**, **Mercer JA**. *Molecular mechanisms and structural features of cardiomyopathy-causing troponin T mutants in the tropomyosin overlap region.* PNAS. 2017;114(42):11115-11120. PMC5651771.
- **Gupte TM**, **Haque F**, **Gangadharan B**, **Sunitha MS**, **Mukherjee S**, **Anandhan S**, **Rani DS**, **Mukundan N**, **Jambekar A**, **Thangaraj K**, **Sowdhamini R**, **Sommese RF**, **Nag S**, **Spudich JA**, **Mercer JA**. *Mechanistic heterogeneity in contractile properties of alpha-tropomyosin (TPM1) mutants associated with inherited cardiomyopathies.* Journal of Biological Chemistry. 2015;290(11):7003-7015. PMC4358124.
- **Sahu N**, **Baligar P**, **Midha S**, **Kundu B**, **Bhattacharjee M**, **Mukherjee S**, **Maushart F**, **Das S**, **Loparic M**, **Kundu SC**, **Ghosh S**, **Mukhopadhyay A**. *Nonmulberry silk fibroin scaffold shows superior osteoconductivity than mulberry silk fibroin in calvarial bone regeneration.* Advanced Healthcare Materials. 2015;4(11):1709-1721.

Patient-derived iPSC line reports

Genomic quality control contributed from the Stanford Cardiovascular Institute iPSC biobank pipeline.

- **Chörsi MS**, **Fernandez HM**, **Tan R**, **Mukherjee S**, **Spiekerkoetter E**, **Wu JC**. *Generation of two induced pluripotent stem cell lines from hereditary hemorrhagic telangiectasia patients harboring ACVRL1 mutations.* Stem Cell Research. 2026;96:104095.
- **Patra D**, **Cabrera DGT**, **Yang X**, **Walther P**, **Mukherjee S**, **Sampson J**, **Wu JC**. *Generation of two induced pluripotent stem cell lines from Fabry disease patients carrying GLA mutations.* Stem Cell Research. 2026;95:104088. PMC13618801.
- **Liu TD**, **Sun S**, **Ren L**, **Zhuge Y**, **Mukherjee S**, **Liang DH**, **Wu JC**. *Generation of induced pluripotent stem cell lines from patients with thoracic aortic aneurysm carrying the ACTA2 gene mutation.* Stem Cell Research. 2025;88:103833. PMC12767169.

PLATFORMS AND PROJECTS

- ADMET-AI. Open ADMET prediction platform, among the fastest and most accurate freely available. Greenstone Biosciences with the Zou group at Stanford.
- ADMET-AI DICT. Cardiotoxicity predictor trained on an FDA dataset, with interpretability analysis of the properties driving each prediction.
- DiGePred. Random forest framework for digenic disease gene pairs, applied across Undiagnosed Diseases Network patients and rare disease cohorts.
- Kinase knowledge graph and research assistant. In development. A locally hosted language model plans a question, consults specialist tools over a human kinase graph built from ChEMBL, BindingDB, profiling panels, structures, trials and patents, and cites the rows behind each claim. Next stage adds Enamine REAL design space and multi-parameter optimization.
- Human pan-fibrosis atlas. In development. Organ-by-organ meta-analysis of public human RNA-seq to find a shared fibrotic program and its targets, with LINCS perturbation signatures ranking compounds that reverse the signature.
- Calcium imaging pipeline for iPSC cardiomyocytes. In development. Segments organoids in 96-well plates and measures beat rate, transient duration, kinetics, beat-to-beat variability and synchrony, with vehicle normalization for drug response and arrhythmia.
- Agentic biomedical research platform. Seven-stage pipeline in Python, FastAPI and PostgreSQL covering entity resolution, intent classification, planning, evidence retrieval across PubMed, GEO, ClinicalTrials.gov, ChEMBL, ClinVar and Reactome, ranking, extraction and aggregation. 77 tests run against a fake model client with no network access.
- SLURM whole-genome sequencing pipeline with multi-lane input, GATK4 joint genotyping, VEP annotation, coverage-mode branching and resume after failure.

TECHNICAL SKILLS

Machine learning and AI: PyTorch, TensorFlow, scikit-learn, XGBoost, random forests, SHAP and model interpretability, feature engineering, active learning, generative models for small molecules, language model application development.

Cheminformatics and drug design: RDKit, QSAR, SAR modeling, ADMET modeling, multi-parameter optimization, structure-based design, molecular docking with AutoDock Vina, SyntheMol and Enamine REAL enumeration, tautomer handling, kinase panel and kinome selectivity profiling, off-target prediction.

Molecular simulation: molecular dynamics, enhanced sampling, free-energy analysis, conformational dynamics. GROMACS, OpenMM, NAMD, PyMOL.

Computational biology and omics: bulk RNA-seq with DESeq2, single-cell analysis with Scanpy and Seurat, pseudobulk methods, whole-genome sequencing with GATK4 and VEP, proteomics, multi-omics integration, network biology, biomarker discovery, variant interpretation.

Programming and infrastructure: Python, R, SQL, Bash, Linux, HPC and SLURM, Git, FastAPI, PostgreSQL, Pydantic, Kuzu, reproducible pipelines, JavaScript and HTML.

Communication and tooling: Streamlit dashboards, interactive browser consoles, matplotlib, seaborn, Plotly, scientific writing, cross-functional communication, mentoring.

Regulatory context: New Approach Methodologies, the FDA Modernization Act 2.0, and iPSC-based human-relevant models as alternatives to animal testing.

HONORS AND FELLOWSHIPS

Vanderbilt International Student and Scholar Program Graduate Student Fellowship, Vanderbilt University, 2016 to 2018

CSIR-NET Research Fellowship, All India Rank 51, Council of Scientific and Industrial Research, India, 2013 to 2016

LEADERSHIP AND SERVICE

Vanderbilt Graduate Student Council, 2018 to 2019. Planned networking events for the whole graduate student population, drawing up to 500 attendees.

Vanderbilt Biological Sciences Graduate Student Association, 2018 to 2019.

Vanderbilt International Biomedical Student Association, 2017 to 2018.

Dramchandran's Lot, University of Calcutta, 2011 to 2013. Founding member of the theatre society, which wrote and staged original plays using cell biology as a metaphor for social questions.

LANGUAGES

English (full professional), Bengali (native), Hindi (professional working), German (elementary).